EXECUTIVE BRIEF

Trusted context. Clear next steps.

One longitudinal care note that turns fragmented clinical, staff and AI entries into a source-linked consult glance - while keeping authorization, revision history and clinician judgment explicit.

11.68 ms
WARM SSR P95 / TARGET <= 300 MS

16 + 1
DOMAIN TESTS + SHELL CHECK

4 / 4
PRIORITY CARDS RESOLVE TO A SOURCE

Highest-value trust loop

01 **GLANCE** — Four capped, ranked cards →
02 **VERIFY** — Exact quote + version →
03 **DECIDE** — Confirm, reject or edit →
04 **ABSTAIN** — Block, review + audit

Runtime architecture

ROLE-SAFE UI — server-filtered bundle →
API GUARD — role + clinic + owner →
D1 RECORD — entries + versions →
TRUST LEDGER — provenance + audit

AI BOUNDARY

Voice/text intake -> measured redaction gate -> strict extraction schema -> local quote/offset verifier -> pending-review entry.

No external model call runs. Generated patient instructions are a separate type and remain blocked until conflict resolution plus clinician approval.

72-hour product decision

Build one complete, inspectable trust loop before adding real ambient capture or broad EHR scope. Every record and identity in the demo is synthetic.

SERVER-ENFORCED

Roles, evidence and safe abstention

UI visibility is never authorization. Each protected request derives the actor from an HttpOnly session and rechecks role, clinic and section ownership.

RBAC matrix

ACTOR	READ	WRITE
Patient	Patient-visible instructions only	No internal notes or comments
Staff	Same-clinic staff/patient/shared AI	Staff notes + comments
Clinician	All same-clinic entries + AI context	Clinician notes + AI review
Admin	Same-clinic oversight + audit metadata	No clinical/staff overwrite

Three distinct accountability records

REVISION

Recoverable content snapshot

full content + version + editor

PROVENANCE

How an output came to exist

source + version + exact quote

CONFLICT/AUDIT

What disagreed; who resolved it

two sources + outcome; no overwrite

Evaluation contract: meaning -> error test -> action

1. Evidence = verified claims / total claims; broken span -> remove or abstain
2. Risk score has deterministic floors; allergy never falls below critical
3. Redaction tests identifier recall and clinical-fact preservation; leak -> fail closed
4. Patient generation needs sources + no open conflict + clinician approval

LEARNING BOUNDARY

Eligible non-critical acceptance adds a bounded +4. Rejection is recorded but does not auto-demote; critical classes ignore learned down-rank. A shadow holdout remains the next exposure-bias gate.

MEASURED AND EXPLICIT

Evidence, performance and next steps

4.76 ms	11.68 ms	12.50 ms
WARM SSR P50	WARM SSR P95	WARM SSR MAXIMUM

5 warm-ups + 30 measured sequential requests against the built Worker. This is shell time, not an internet or clinical SLA.

Automated validation

- Clinic and role scope; patient internal-note boundary
- Immutable revisions; revert creates a new version
- Exact source/version/span; fabricated anchor is rejected
- Stale same-section write returns deterministic conflict
- Separate role-owned sections do not overwrite each other
- Critical floor ignores feedback; rejection is not learned
- Human-human dose conflict blocks patient release
- Identifiers removed while dose/medication facts survive

72-hour scope: now vs. next

BUILT NOW

- Glance + exact-span evidence
- Risk floor + patient abstention
- Version/edit + clinical conflicts
- Bounded feedback + audit
- Measured pre-model redaction gate

NEXT VALIDATION GATE

- Clinic SSO and membership claims
- Shadow holdout + calibrated eval
- Broader terminology conflict rules
- Stronger entity detection + review
- Security/legal review before real data

Evidence used to shape the build

Singapore MOH AIHGle 2.0 - human oversight, source checks and drift monitoring

NIST AI 600-1 + Measure - ground truth, lineage and use-context evaluation

HL7 FHIR R5 - Provenance and AuditEvent separation

JAMIA selective prediction - defer when error cost exceeds review cost

OpenAI - Structured Outputs, evaluation guidance and API data controls

OWASP A01 - server-side, deny-by-default access control

Full URLs and claim-level notes: README.md and research/claim-source-ledger.md